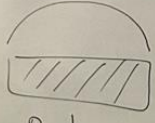
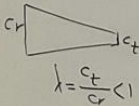


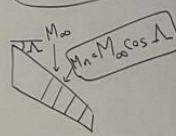
Compressible flow where $\rho \neq \text{cte}$
occurs when $M = \frac{u}{a} > 0.3$



Rectangular
(heavy)



$k = \frac{t}{c} < 1$



$M_{\infty} \uparrow$ shock is stronger
 $M_{1n} = M_{\infty} \cos \Delta$

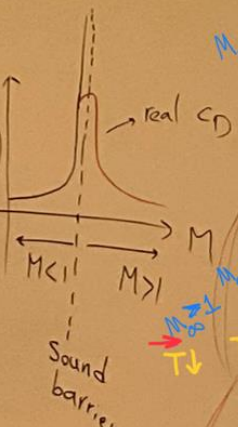
$\rho_{\infty} = \frac{P_{\infty}}{RT_{\infty}}$

$V_{\infty}^2 = M_{\infty}^2 \gamma R T_{\infty}$

$D = C_D \frac{1}{2} \rho_{\infty} V_{\infty}^2 S$

$D = C_D \frac{\gamma}{2} P_{\infty} M_{\infty}^2 S$

$L = C_L \frac{\gamma}{2} P_{\infty} M_{\infty}^2 S$



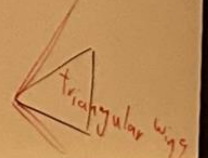
real C_D

$M < 1$ $M > 1$

Sound barrier

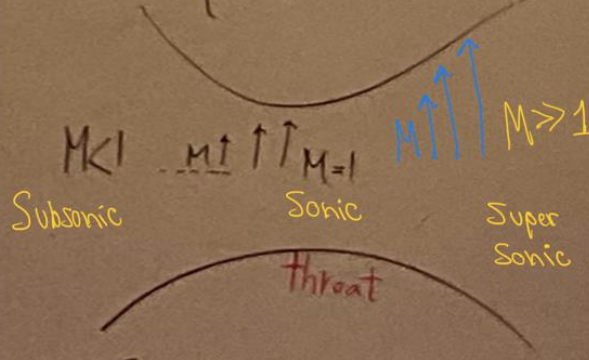


$M_{\infty} \uparrow$ $M_{\infty} \uparrow$
Diamond shape



$M_{\infty} \uparrow$ $M_{\infty} \uparrow$
Triangular wings

Subsonic $M \uparrow$ as $A \downarrow$
Supersonic $M \uparrow$ as $A \uparrow$



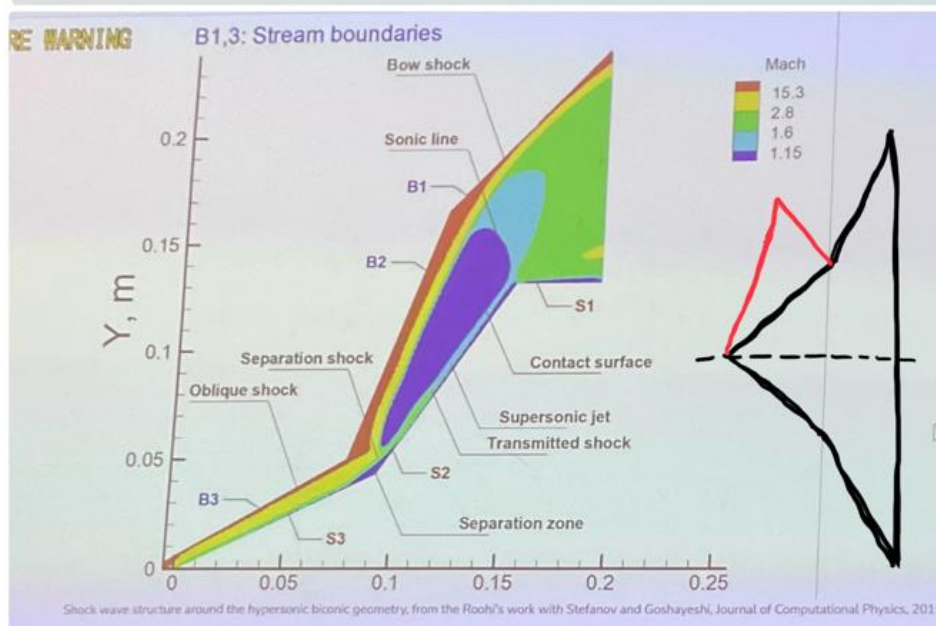
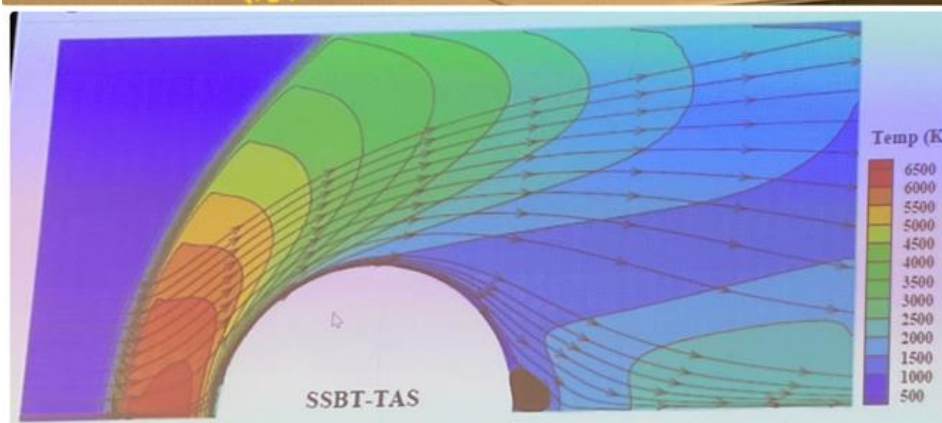
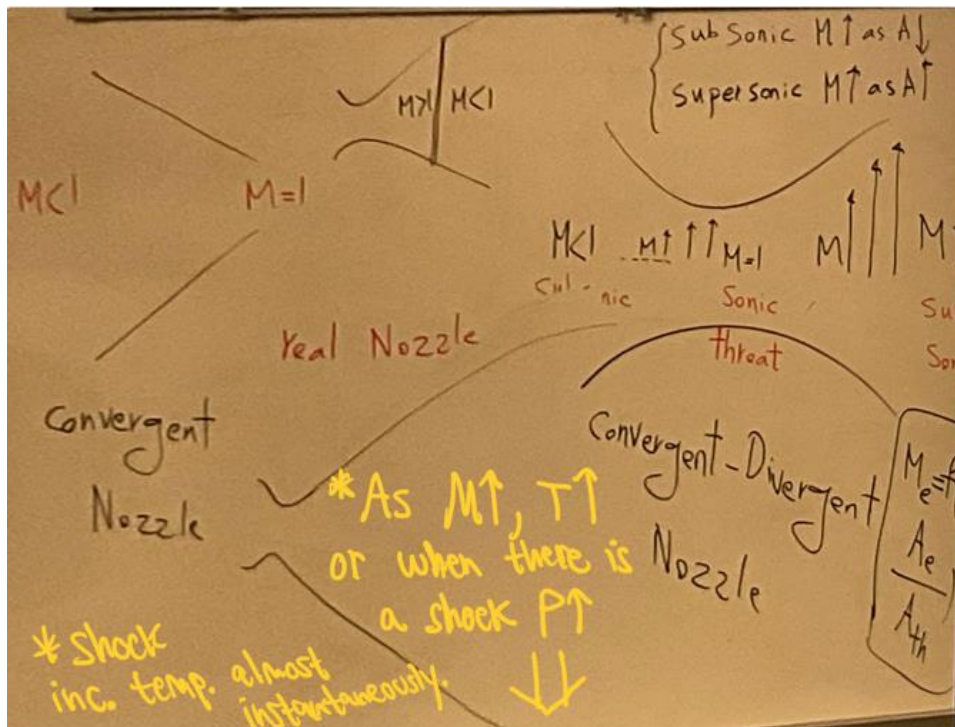
$M < 1$ $M \uparrow \uparrow \uparrow M = 1$ $M \uparrow \uparrow \uparrow M \gg 1$

Subsonic Sonic Super Sonic

Throat

Convergent-Divergent
Nozzle

$M_{\text{exit}} = f\left(\frac{A_e}{A_{th}}\right)$



(SEE NOTES POSTED)

Some Introductory Concepts [8.1-8.3]

Objective: Review of basic thermodynamic concepts (including 2nd law); develop isentropic equations

Brief Review of Thermodynamics [8.1]

$$p = \rho R T$$

General gas $R = \frac{\bar{R}}{M}$

$$\bar{R} = 8314 \frac{\text{J}}{\text{kmol} \cdot \text{K}}$$

M : Molar mass

p : pressure

ρ : density

T : absolute temperature

$$\underline{p = \rho R T}$$

$R = \frac{\bar{R}}{M} \rightarrow$ universal gas constant (1.10) $\frac{\text{J}}{\text{kg} \cdot \text{K}}$
 $\bar{R} \rightarrow$ mass of one mole $\frac{\text{J}}{\text{kmol} \cdot \text{K}}$
 $M \rightarrow$ mass of one mole $\frac{\text{kg}}{\text{kmol}}$

$R =$ specific gas constant $= \boxed{287} \frac{\text{J}}{\text{kg} \cdot \text{K}} = 1716 \frac{\text{ft} \cdot \text{lb}}{\text{slug} \cdot \text{R}}$

for air

Or $\underline{p v = R T}$ v specific volume $= \frac{1}{\rho}$

this is accurate (< 1% error) for low pressures ($\leq 1 \text{ atm}$)

moderate temperatures ($\geq 273 \text{ K}$)

Specific Heats [8.1.1-2]

internal energy: $\underline{u_e} \equiv u_e(v, T)$ (8.1)

enthalpy: $\underline{h} \equiv h(T, p) = \underline{e} + pv = u_e + \frac{p}{\rho}$ (8.6)

if the gas is thermally perfect gas: $\underline{u_e} = u_e(T)$, $\underline{h} = h(T)$

$$\underline{de = du_e = c_v dT}, \quad \underline{dh = c_p dT} \quad (8.3, 8.10)$$

$$e = u_e = c_v (T_2 - T_1) \quad h = c_p (T_2 - T_1)$$

for $\underline{c_p}, \underline{c_v}$ constant $\Rightarrow$ calorically perfect gas, but $\begin{cases} c_p \\ c_v \end{cases}$ are function of T

$$e = u_e = c_v T \quad \text{where:}$$

$$c_v = \left(\frac{\partial u_e}{\partial T} \right)_v \quad \text{At low } T, \text{ we assume they are constant (8.3)}$$

$$\underline{h = c_p T}$$

where:

$$c_p = \left(\frac{\partial h}{\partial T} \right)_p \quad c_v = \frac{R}{\gamma - 1} \quad (8.11)$$

$$\underline{c_p - c_v = R}$$

gas
Constant

$$\gamma = \frac{c_p}{c_v} \Rightarrow c_p = \frac{\gamma R}{\gamma - 1} \quad (8.15)$$

$$\gamma = \frac{n+2}{n}$$

n : # of molecular degrees of freedom

= 1.29 for monatomic gases
= 1.4 for diatomic gases

$n=5$
air

3 for mono
 $\gamma = 1.67$
5 for diatomic
 $\gamma = 1.4$

= 1.67 for triatomic gases, eg. CO_2

$n=3$
mono

$$\gamma = \frac{n+2}{n} \Rightarrow 1 - \frac{c_v}{c_p} = \frac{R}{c_p} \Rightarrow 1 - \frac{1}{\gamma} = \frac{R}{c_p} \Rightarrow$$

$$n: \text{degrees of freedom} \Rightarrow c_p = \frac{\gamma R}{\gamma - 1} \text{ and } c_v = \frac{R}{\gamma - 1} \quad (8.16)$$

monatomic gas: can be found for air (R, γ are known)

$\gamma = \frac{n+2}{n}$ → Monatomic gas can translate in 3-directions

(x-y-z) in space, so $\Rightarrow \gamma = \frac{5}{3} = 1.67$

Diatomic gas at low T $\left\{ \begin{array}{l} 3 \text{ translational D.O.F} \\ 2 \text{ rotational D.O.F} \end{array} \right. = 5$

$\gamma = \frac{7}{5} = 1.4$ (Air is the same)
in this class

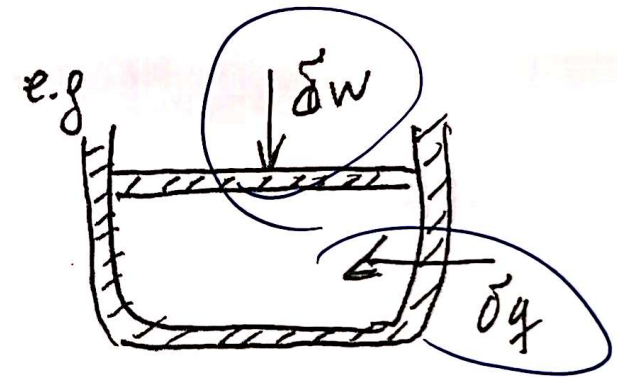
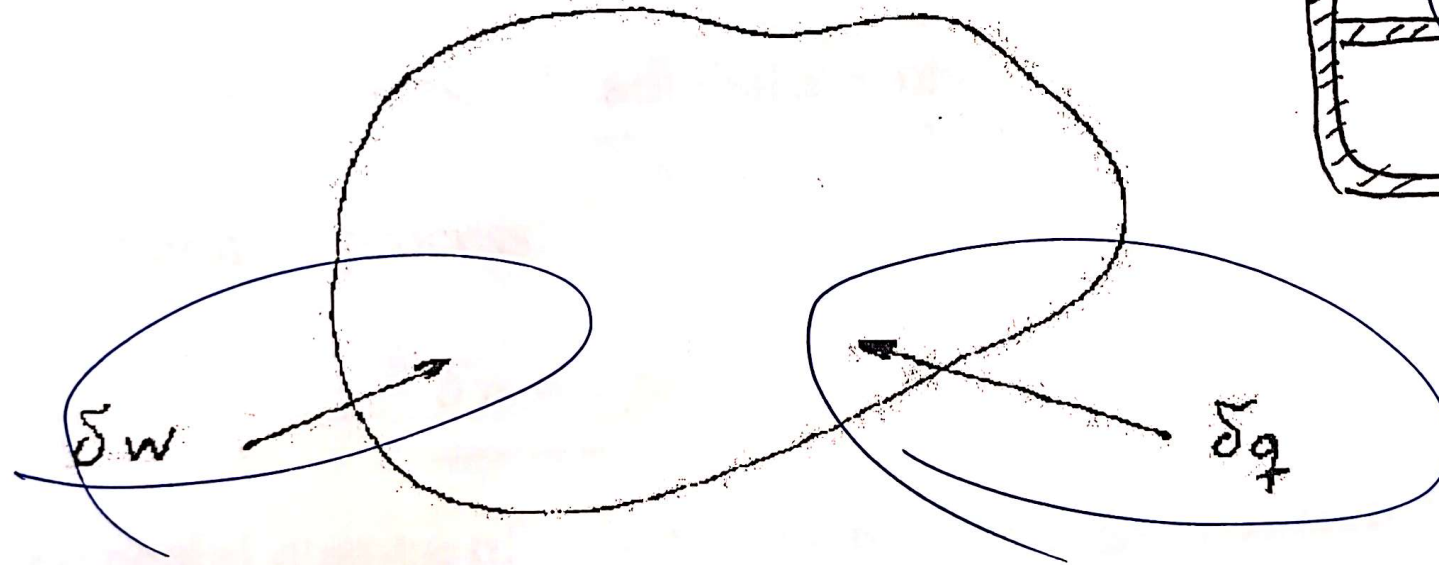
Triatomic → complex degrees of $\left\{ \begin{array}{l} 3 - \text{translational} \\ ? - \text{rotational} \\ ? - \text{vibrational} \end{array} \right.$

CO_2 → usually has $n=7$ so $\gamma = \frac{9}{7} = 1.29$

$$\delta w = -\int p dv$$

First Law of Thermodynamics:

system (fixed mass)



$$\delta q + \delta w = \overbrace{\delta u_e}^{\delta e} = \text{change in energy}$$

δq = heat added, δw = work done, δu_e = change in energy

- *adiabatic* process: no heat added or taken away $\Rightarrow \delta w = \delta u_e$
- *reversible* process: no dissipative phenomena occur $S_{gen} = 0$
- *isentropic* process: adiabatic + reversible

for a reversible process

$$\underline{\delta w = -p dv}$$

incremental change of volume due to boundary displacement

$$\Rightarrow \boxed{\delta q - p dv = \delta u_e} \quad (8.1)$$

(alternative form of 1st law)

Entropy and second law [section 8.1.3]

for an adiabatic process:

$$ds = \int \frac{\delta q}{T} + ds_{\text{gen}} \quad \text{heat transfer}$$

entropy

(8.17)

$q = 0$

$$ds = ds_{\text{gen}} \geq 0$$

First law:

~~you cannot win~~

Second law:

~~you always lose~~

$$ds = \frac{\delta q}{T} + ds_{\text{gen}} \rightarrow \text{reversible process} \Rightarrow \delta q = T ds \rightarrow$$

Calculation of entropy (for a reversible process)

put in first law

$$\Rightarrow \delta q = T ds \Rightarrow \boxed{T ds - \frac{\delta w}{p} = du_e}$$

$$\Rightarrow \boxed{T ds = du_e + p dv}$$

(8.18)

use enthalpy $h = u_e + pv$

$$\Rightarrow dh = \overbrace{du_e + pdv}^{Tds \text{ from 8.18}} + vdp$$

$$\Rightarrow \boxed{Tds = dh - vdp}$$

The above equation and 8.18 are alternative forms of 1st law

for a thermally perfect gas:

$$dh = \underline{c_p dT} \quad (8.10)$$

$$\Rightarrow ds = c_p \frac{dT}{T} - \frac{vdp}{T}$$

assume:

$$\begin{cases} pv = RT & (\text{ideal gas law}) \\ \frac{v}{T} = \frac{R}{P} & (*) \end{cases}$$

$$\begin{cases} (Tds = c_p dT - vdp) / T \\ ds = c_p \frac{dT}{T} - \frac{v}{T} dp \rightarrow \text{use } (*) \end{cases}$$

$$ds = c_p \frac{dT}{T} - \frac{R}{P} dp$$

$$\text{Assume: } c_p \text{ \& } R = \text{cte} \Rightarrow s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$$

$$\Rightarrow ds = c_p \frac{dT}{T} - R \frac{dp}{p} \Rightarrow s_2 - s_1 = \int_{T_1}^{T_2} c_p \frac{dT}{T} - R \ln \frac{p_2}{p_1}$$

for a calorically perfect gas ($c_p = \text{constant}$)

$$\Rightarrow s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \quad (\text{equivalent to 8.19b})$$

similarly: (use $p\nu = RT$, $c_p = \frac{\gamma R}{\gamma - 1}$; $c_v = \frac{R}{\gamma - 1}$ 8.16)

Eq 2

$$\Rightarrow s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{\nu_2}{\nu_1} \quad \nu = 1/\rho \quad (\text{equivalent to 8.19c})$$

useful eqs to evaluate entropy increases

Isentropic relations

put $s_1 = s_2$ in (8.19a)

$$\Rightarrow 0 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \Rightarrow \ln \frac{p_2}{p_1} = \frac{c_p}{R} \ln \frac{T_2}{T_1}$$

$$\Rightarrow \frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{c_p/R}$$

from (8.16) $\frac{c_p}{R} = \frac{\gamma}{\gamma - 1}$

$$\Rightarrow \frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}$$

$$c_p \ln \frac{T_2}{T_1} = R \ln \frac{p_2}{p_1}$$

$$\frac{\cancel{\gamma R}}{\gamma-1} \ln \frac{T_2}{T_1} = \cancel{R} \ln \frac{p_2}{p_1} \Rightarrow \ln \left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}} = \ln \left(\frac{p_2}{p_1} \right)$$

$$\left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}} = \frac{p_2}{p_1}$$

isentropic relation

$$P = \gamma R T \quad (1) \text{ ideal gas}$$

$$\left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}} = \frac{p_2}{p_1} \quad (2) \text{ isentropic gas}$$

similarly:

$$\Rightarrow \frac{\rho_2}{\rho_1} = \left(\frac{T_2}{T_1} \right)^{1/(\gamma-1)}$$

use the other entropy relation
isentropic Eq 2
relations

in summary:

$$\boxed{\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1} \right)^\gamma = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}}$$

or:

use Eq 2

for isentropic flows

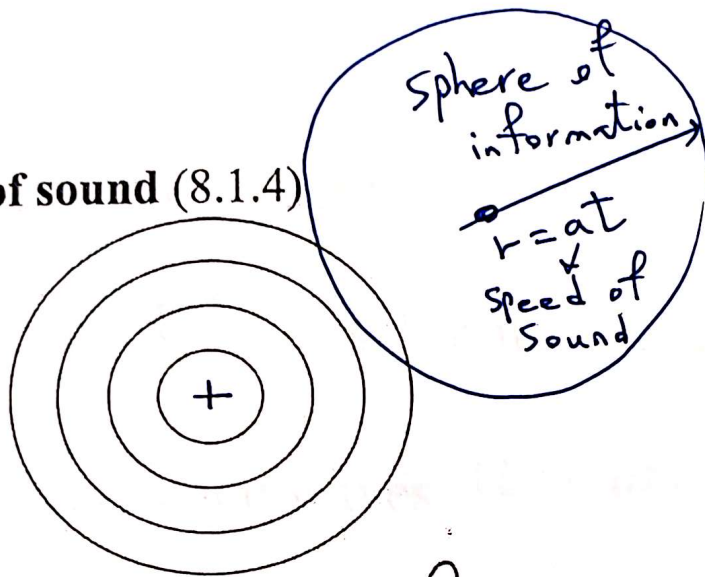
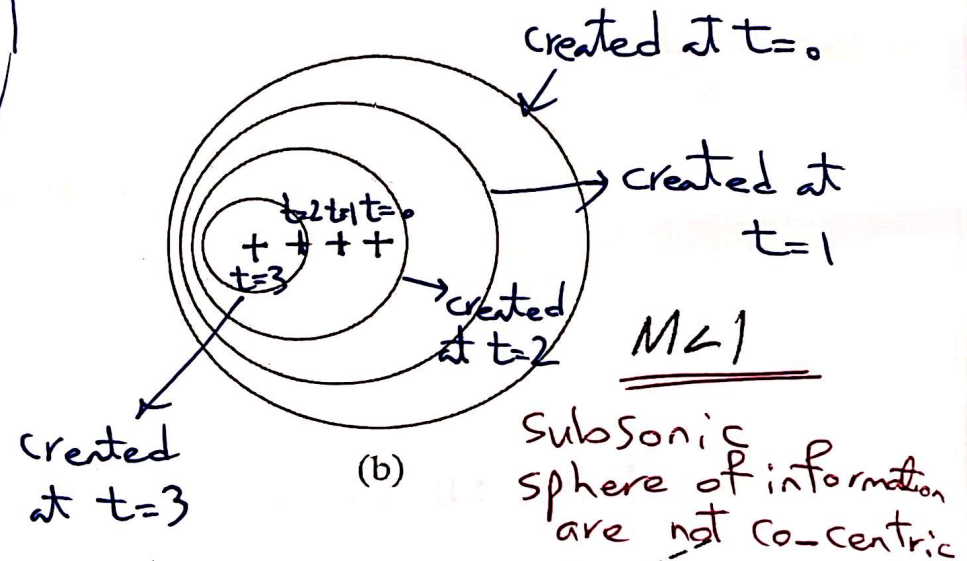
(8.20)

$$\boxed{\frac{p}{\rho^\gamma} = \text{constant}; \quad \frac{T^{\gamma/(\gamma-1)}}{p} = \text{constant}; \quad \frac{T}{\rho^{(\gamma-1)}} = T v^{(\gamma-1)} = \text{constant}}$$

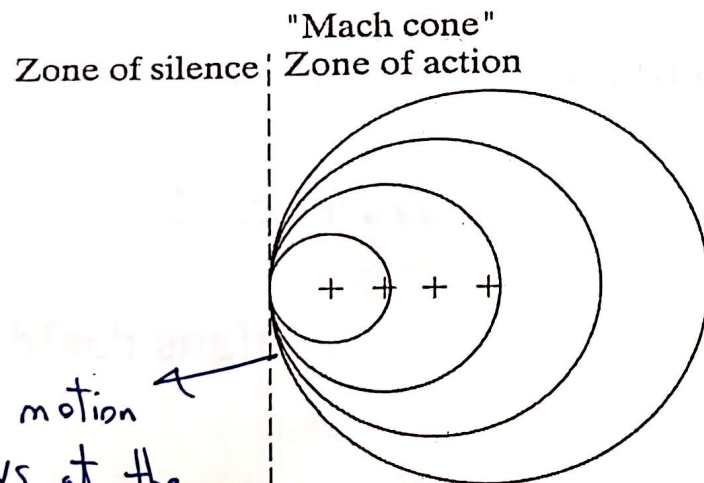
(energy relation for an isentropic process) many processes can be

assumed isentropic.

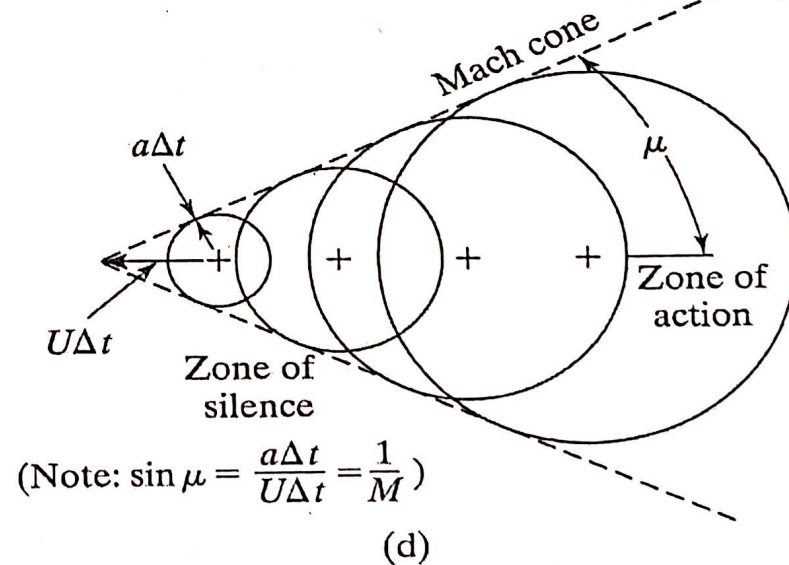
Speed of sound (8.1.4)

(a) $M=0$ 

(b)



(c)

 $M=1$ 

(d)

 $M > 1$

$$(\text{Note: } \sin \mu = \frac{a\Delta t}{U\Delta t} = \frac{1}{M})$$

$M=1$ motion
always at the
edge of sphere